EE366200 DSP Laboratory

Lab 6 Image Filtering and Hybrid Images

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1. Introduction

Hybrid images exploit a simple but powerful fact of human vision: when we stand close to an image our perception is dominated by high spatial frequencies, while from a distance low spatial frequencies prevail. This project studies that phenomenon through hands-on image filtering and synthesis. First, I implement a general two-dimensional convolution routine capable of applying a variety of kernels to grayscale and color images, which serves as the foundation for constructing frequency-selective transforms. Building on this, I generate hybrid images by low-pass filtering one source with a Gaussian blur and high-pass filtering another by subtracting a blurred version from its original, then summing the two results so that one subject emerges at near viewing and the other at far viewing. I further examine filter choice and parameterization by contrasting Gaussian blur with box averaging and by observing derivative operators such as Sobel and Laplacian, including an alternative pipeline that approximates the low- and high-frequency components using the Laplacian. Because convincing hybrids require structural correspondence, I also align challenging pairs—

particularly faces—via geometric normalization before blending, and I analyze the visual outcomes, artifacts, and limitations that follow from these design decisions.

2. Objectives

The goal of this work is to understand and demonstrate how spatial frequency content governs human perception by creating images that convey different identities at different viewing distances. To that end, I aim to build a reliable image filtering capability that enables low-pass and high-pass decomposition on both grayscale and color data, and then use these components to synthesize hybrid images in which the low frequencies of one source dominate from afar while the high frequencies of another source dominate up close. I seek to characterize how design choices—such as blur type and strength, kernel size, and band-selection thresholds—affect recognizability, sharpness, and artifacts, and to compare Gaussian smoothing with simpler alternatives and with derivative-based operators conceptually. I also intend to evaluate an alternative formulation that substitutes a Laplacian-based separation for the standard Gaussian pipeline and to determine when this variant helps or harms the perceptual illusion. Because structural correspondence is critical, another objective is to assess the role of geometric alignment on the success of the hybrids and to document the kinds of failures that arise when alignment is inadequate. Finally, I plan to present clear visual evidence and concise analysis that together substantiate these observations and distill practical guidelines for creating convincing hybrid images.

3. Approach

I began by implementing a general two-dimensional convolution function, “my_imfilter”, that accepts grayscale or RGB inputs and any odd-sized kernel. To preserve the input resolution, I used zero padding along the height and width determined by half the kernel dimensions, and I applied the kernel channel-wise so that color structure is filtered consistently. I avoided all built-in convolution routines and instead computed each output pixel as the inner product between the kernel and the corresponding padded window. To verify correctness and numerical behavior, I ran the provided test script on a representative RGB image: the identity kernel confirmed that values are preserved, the small box kernel demonstrated uniform smoothing, and the Sobel and Laplacian derivative kernels produced signed responses that I visualized by adding a constant offset before clipping so that negative values were visible. I also formed a simple high-pass image by subtracting a blurred version from the original to check that edges and fine details were emphasized while flat regions remained near zero. These checks established that padding, channel handling, and value ranges behaved as intended for subsequent experiments. The check results are shown from Fig.1 to Fig.8.

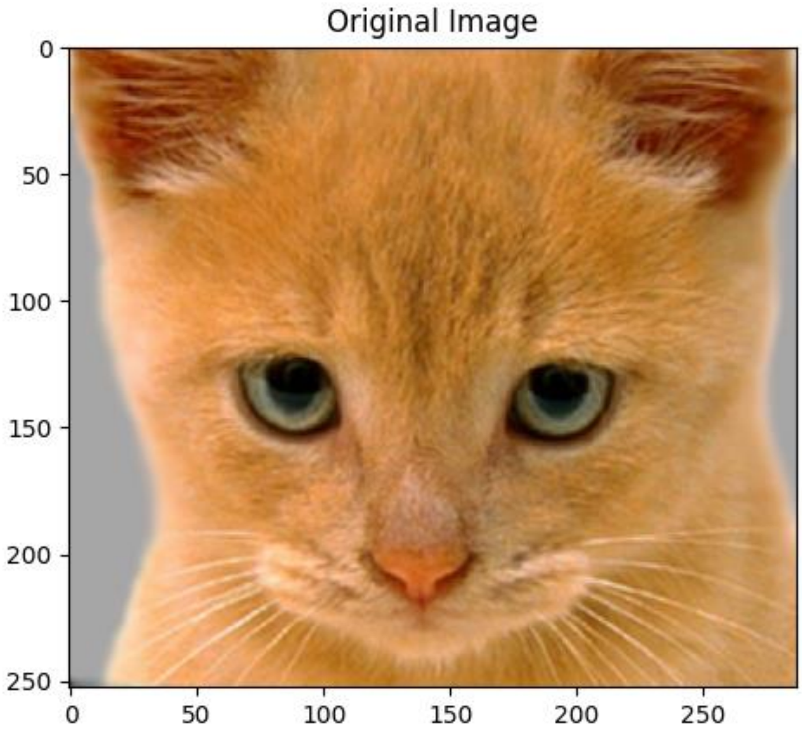

Fig.1 The original image for testing the functionality of “my_imfilter”

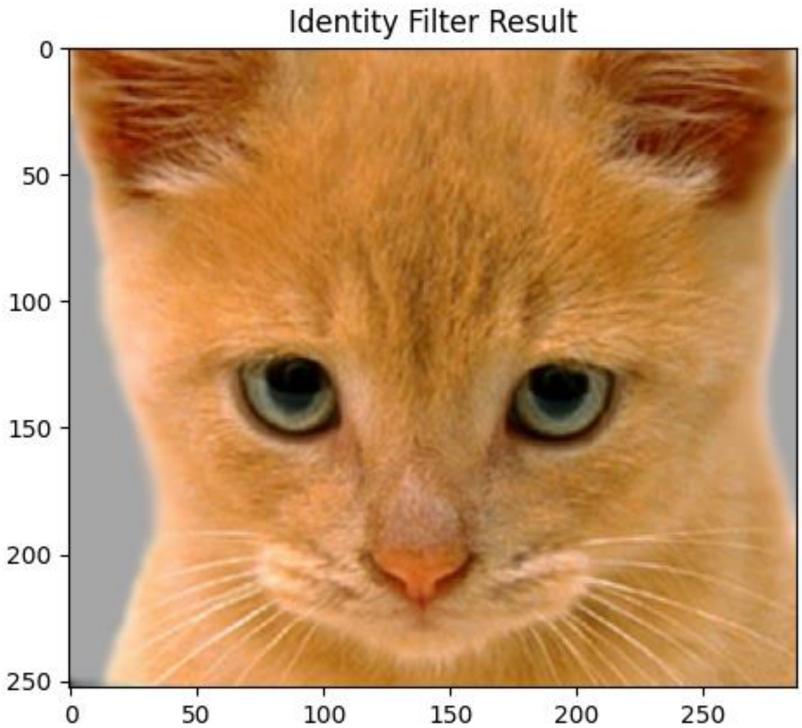

Fig.2 The result of applying an identity filter to the original image

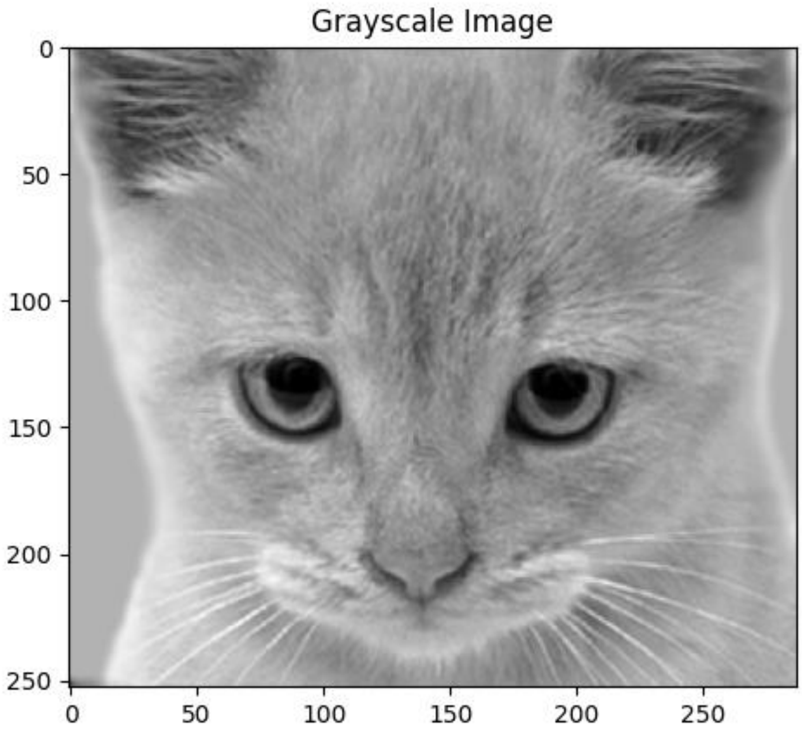

Fig.3 The result of converting the original image into gray scale

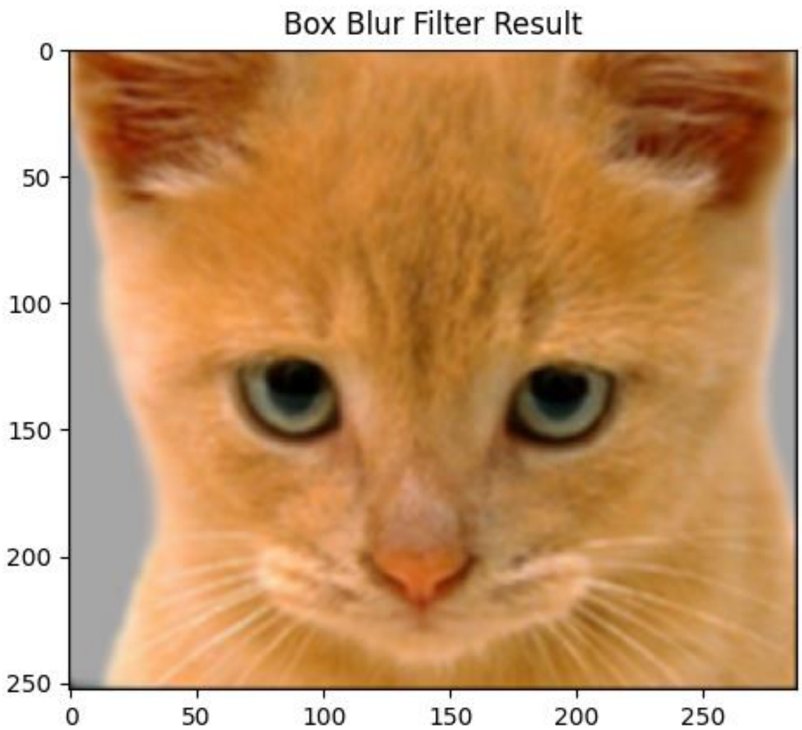

Fig.4 The result of applying a box blur filter to the original image

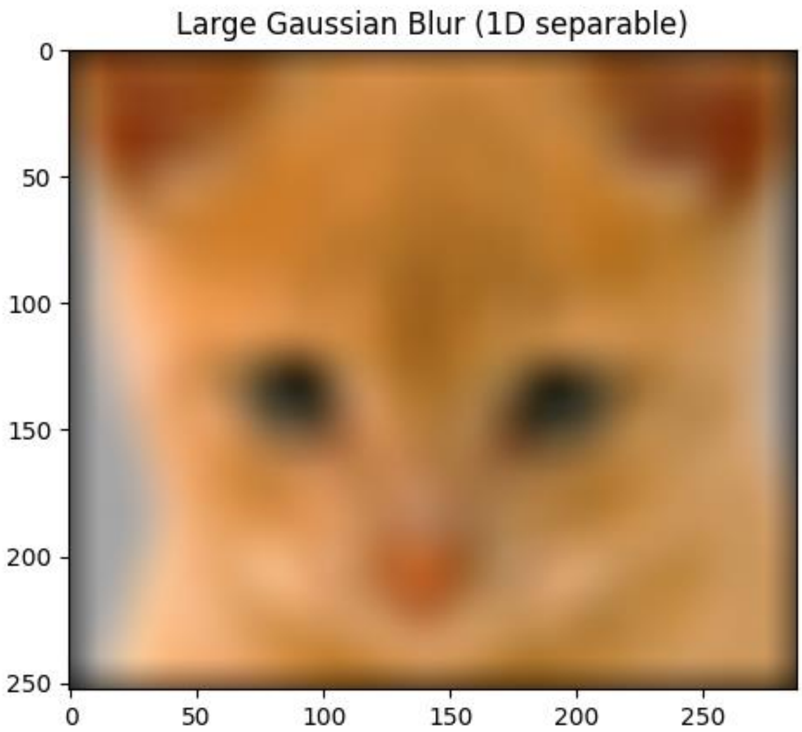

Fig.5 The result of applying a large Gaussian blur filter to the original image

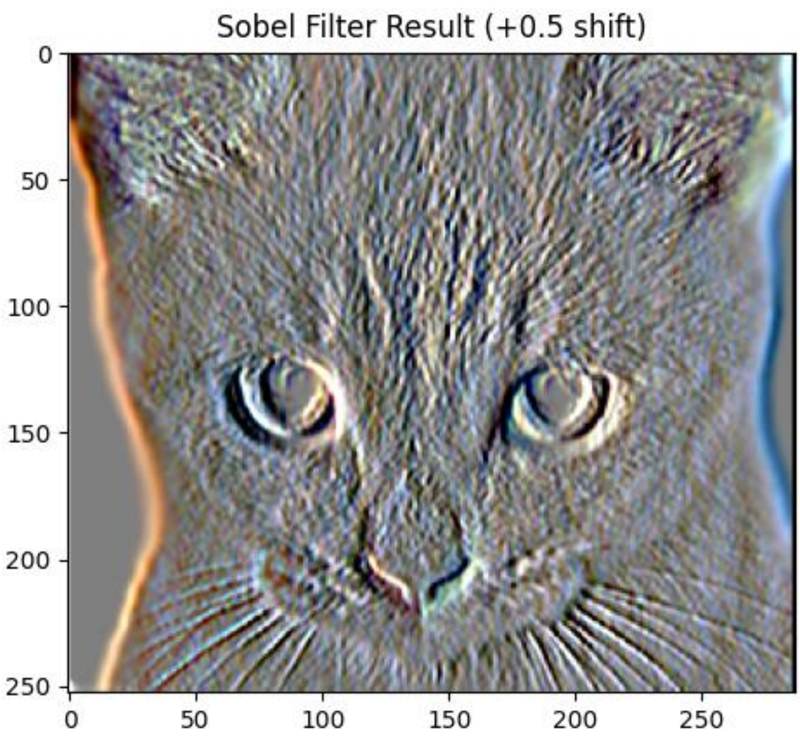

Fig.6 The result of applying a Sobel filter and a +0.5 shift to the original image

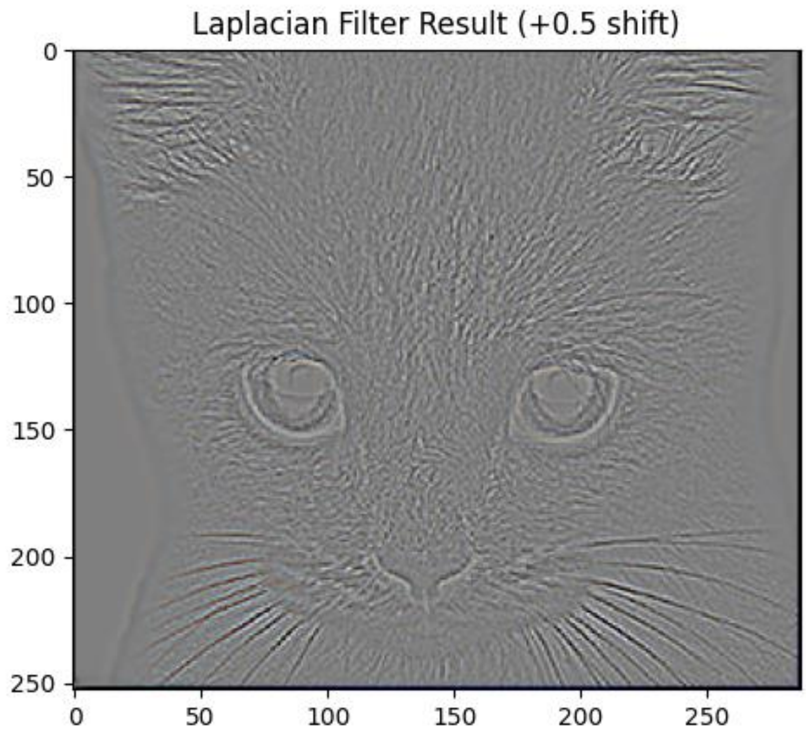

Fig.7 The result of applying a Laplacian filter and a +0.5 shift to the original image

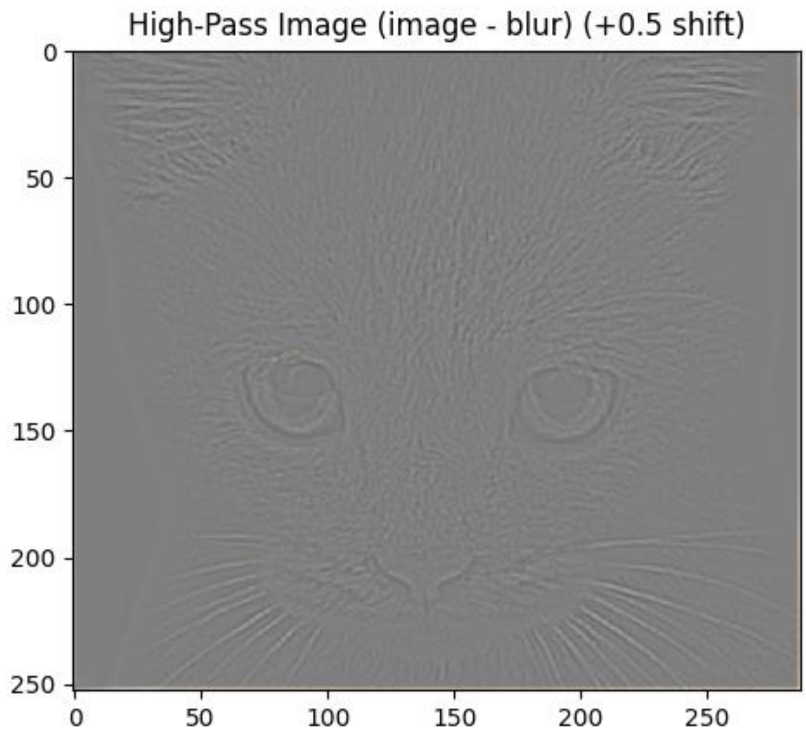

Fig.8 The result of applying a high-pass filter and a +0.5 shift to the original image

With the filtering primitive verified, I completed the second demo by constructing hybrid images as described in the lab, using a Gaussian blur with a tunable standard deviation σ to act as the cutoff-frequency control. I first prepared each pair by converting both sources to floating-point RGB in the [0, 1] range and resizing them to the same spatial resolution so that subsequent filtering and summation were well defined. I produced the low-pass component by convolving the first image with an isotropic Gaussian whose σ I set per pair; the kernel size was chosen large enough relative to σ to avoid truncation artifacts. I produced the complementary high-pass component by subtracting a Gaussian-blurred version of the second image (using the same σ as in the starter flow) from the original, yielding a zero-mean signal of fine detail and edges. I then added the two components to synthesize the hybrid and clipped the result back to the displayable range for saving and visualization. To present intermediate results as suggested, I visualized the high-pass image by adding an offset of 0.5 solely for display so that negative values could be seen, and I created the multi-scale view by progressively down-scaling the hybrid with the provided visualization routine, which makes the shift from high-frequency dominance at close viewing to low-frequency dominance at distance apparent in a single figure. The visualization of the hybrid images are shown from Fig.9 to Fig.11.

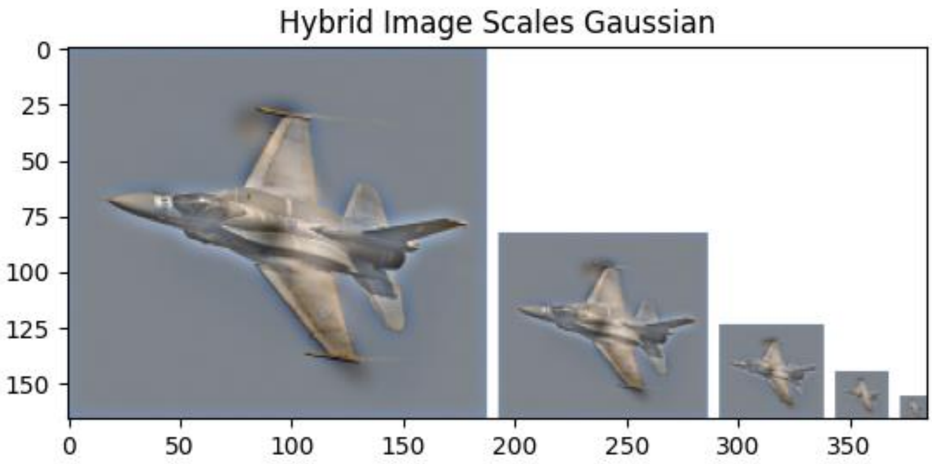

Fig.9 Hybrid image generated using a Gaussian filter with $\sigma = 4$

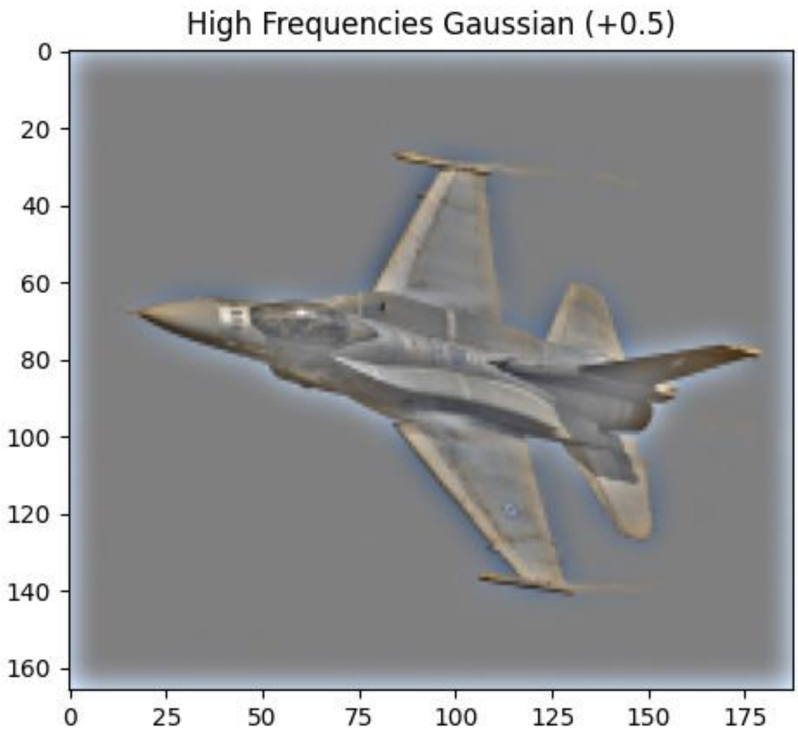

Fig.10 The high frequency component of the hybrid image.

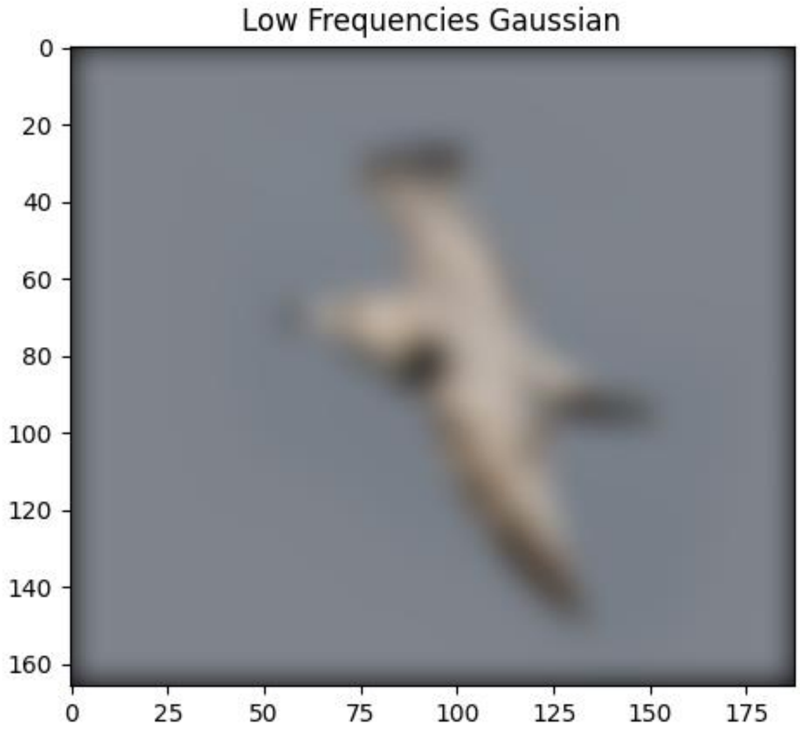

Fig.11 The low frequency component of the hybrid image.

4. Discussion

I. Explain why the high frequency images will be seen at close distance (larger one) while the low frequency images will be seen at far distance (smaller one)?

Ans:

High spatial frequencies in an image correspond to rapid intensity changes—fine edges, small textures, and sharp details—whereas low spatial frequencies represent slow variations—the broad “blob-like” structure and coarse shading that define overall shape and layout. At close viewing distances, the image subtends a larger visual angle, so those fine edge patterns occupy more degrees of visual angle and fall into a range where the human visual system can resolve and attend to them; as a result, the high-frequency component dominates what we perceive up close. As we move farther away, the same physical image shrinks on the retina, effectively shifting its spatial content to higher cycles per degree; high-frequency detail then drops below visibility (or is treated like noise), leaving the coarse, low-frequency structure to determine the percept at a distance. The hybrid-images paper formalizes both sides of this explanation: hybrids are explicitly constructed by adding a low-pass image intended "to be seen at a far distance" and a high-pass image intended "to be seen up close," with the viewing distance at which each dominates governed by the image size and the chosen cut-off frequencies (specified in cycles/image) $\left(H = I_1 \cdot G_1 + I_2 \cdot (1 - G_2)\right)$. The paper also notes why distance matters perceptually: image interpretation is multiscale and proceeds coarse-to-fine, with low spatial frequencies dominating early processing in the visual system; for a given viewing distance, a particular frequency band will dominate perception and stepping back a few meters reliably flips the perceived identity in a hybrid image. Finally, the authors emphasize the units link between filtering and distance: cutoffs are set in cycles/image (distance-independent for a fixed pixel image), while cycles/degree describe what the eye sees at a particular size and distance—so changing distance changes which band remains visible and thus which interpretation win.

II. Discuss different low pass filters (box filter and Gaussian filter under the same size). Which one generates visually-pleasing hybrid images? Please show some examples and try to explain the results with the respect to signal and system.

Ans:

Using the bird–plane pair, I compared Gaussian and box low-pass filters under the same kernel size (the box kernel was explicitly built to match the Gaussian size in the starter flow) and then formed hybrids by adding the bird's low frequencies to the plane's high frequencies. To establish the low-pass behavior, I first present the Gaussian-blurred bird next to the box-blurred bird and describe the spatial artifacts I observed.

Fig.12 The Gaussian-blurred bird (left) and the box-blurred bird (right)

The Gaussian yields a smooth, natural silhouette with soft gradients, while the box blur shows slight "shadow" bands and blocky plateaus near strong edges—a direct consequence of the box filter's sinc-like frequency response with sidelobes (passband ripple and stopband leakage), versus the Gaussian's monotone roll-off without sidelobes.

Because the high-pass image is computed as original minus low-pass, any ripple or leakage from the low-pass appears as halos in the high-pass. To make this visible, I show the airplane's high-frequency component for both case.

Fig.13 The Gaussian-High-Pass airplane (left) and the box-High-Pass airplane (right)

Here, Fig.13 was displayed with a +0.5 offset for visibility. The box-based result exhibits brighter fringes hugging leading edges and tips, whereas the Gaussian-based high-pass isolates edges and fine texture with fewer halos and less spurious mid-band content.

I then show the final hybrids constructed from each low-pass/high-pass pair. Up close, the Gaussian hybrid reads crisply as the plane without glowing contours; as the viewing scale decreases, the bird emerges cleanly from the coarse structure. The box hybrid, by contrast, carries a soft "aura" around high-contrast contours at near distance and lets too much mid-frequency energy bleed into the far-distance view, so the identity switch is less clean.

Fig.14 The Gaussian hybrid image (left) and the box hybrid image (right)

Finally, to make the near-to-far transition explicit in a single figure, I include the multi-scale visualizations here.

Fig.15 The multi-scale visualizations of Gaussian hybrid image

Fig.16 The multi-scale visualizations of Box hybrid image

The Gaussian series shows a smooth perceptual flip with scale, while the box series preserves distracting mid-band remnants during downscaling. From a signals-and-systems perspective, both filters are LTI with unit DC gain, but the Gaussian's smoothly decaying spectrum offers better stopband attenuation and minimal ringing in space, which is exactly what hybrid construction needs: a truly low-frequency base and a clean edge-only complement. Under the same kernel size, the Gaussian therefore produces the more visually pleasing hybrid for this pair.

III. Reverse the flow when generating low-passed filtered image and high-passed filtered image. Specifically, low pass filtered image is obtained by subtracting high pass filtered image. High pass filter is provided below. (Put your code in Lab06/code/ and name it reverse_generate_hybrid_image.py) Please discuss the difference between this reversed flow and flow with Gaussian filter used in this lab. (e.g. quality of hybrid image and low-pass filtered image)

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Ans:

I began by reasoning from the operator itself: the 3×3 Laplacian is a discrete second derivative with zero DC response, so it boosts rapid intensity changes and is very sensitive to noise; around any step edge it produces a bright/dark biphasic outline. If I subtract the Laplacian from the original image, I am not creating a true low-pass; I am performing a high-boost sharpening (a variant of unsharp masking).

If, instead, I subtract the original from the Laplacian, I only flip the edge polarity; the result is still dominated by high frequencies and remains a poor “base” for hybrids.

Following that understanding, I implemented the reversed flow exactly on the bird–plane pair: I defined the “low-pass” as original − Laplacian(original) from the bird, and the high-pass as Laplacian(original) from the plane. Because the raw Laplacian magnitude is much smaller than the base layer, I deliberately changed the plain sum into a weighted sum to make the mixed percept readable; I used base + 5×high so that the plane’s edge structure is visible without being swamped by the base.

Fig.17 Low-frequency component of the hybrid image using Laplacian filter

Instead of a smooth, blob-like bird, the frame already looks sharpened: the feather contours and wing edges are accentuated, and there is fine speckle even in the background. This confirms that my “low-pass” layer leaks high-frequency energy and cannot serve as a clean coarse component.

The complementary edge map is visible in Fig.18

Fig.18 High-frequency component of the hybrid image using Laplacian filter

The aircraft's leading edges, canopy seams, and wing tips appear as bright/dark outlines with thin halos. I also see faint colored speckles along some contours—an expected artifact when a second derivative is applied independently to RGB channels and then recombined.

I then formed the hybrid with the weighted sum

Fig.19 hybrid image using Laplacian filter

At close range the airplane is certainly readable, but halos trace most high-contrast edges and a fine "sand" texture covers the background and fuselage.

Comparing this to the Gaussian-based hybrids I built elsewhere in the lab, the difference is clear: the Gaussian pipeline delivers a genuinely smooth low-frequency base and a clean edge-only high-pass, so the near-to-far identity switch happens cleanly as scale decreases. In the reversed Laplacian flow, the base itself is sharpened, so edge energy persists even when I step back; the hybrid looks busier at intermediate scales and the perceptual flip is less decisive. That is why, to present a better mixed percept, I converted the direct addition into a weighted addition (base + 5×high), which improves near-view readability but cannot fully remove the halos or mid-band leakage caused by defining the "low-pass" as original − Laplacian(original).

IV. Try different settings of the sigma (e.g. 3, 7, 11…) in "generate_hybrid_image.py". Show your results and explain what you observe (e.g. differences between the results of different sigma, execution time, filter kernel size, blurred quality…). Why does changing sigma cause such differences?

Ans:

Starting with the edge channel, Fig.20 shows the high-pass results (displayed with a +0.5 offset). At $\sigma = 0$ the image is uniform mid-gray because "high = original − blur" vanishes when blur is the identity. At $\sigma = 1{\sim}2$ the aircraft appears as very thin, crisp contours: only the highest spatial frequencies survive, so shading is almost absent. At $\sigma = 3$ the contours thicken slightly and just enough mid-band detail appears on the fuselage and wings; halos remain mild and the background is quiet. As σ increases from 4 to 6, edge lobes broaden and a soft halo forms around high-contrast boundaries; background structures begin to creep in, especially near the frame edges. From $\sigma = 7$ to 14 the high-pass turns into a strong “embossed” relief: edges are wide and bright, mid-band textures remain, and the border glow is obvious because the Gaussian kernel has become large. This monotonic thickening is the expected effect of lowering the low-pass cutoff by increasing σ: the complementary high-pass $1 - H_\sigma$ no longer isolates only the very high frequencies but also retains more of the mid-band.

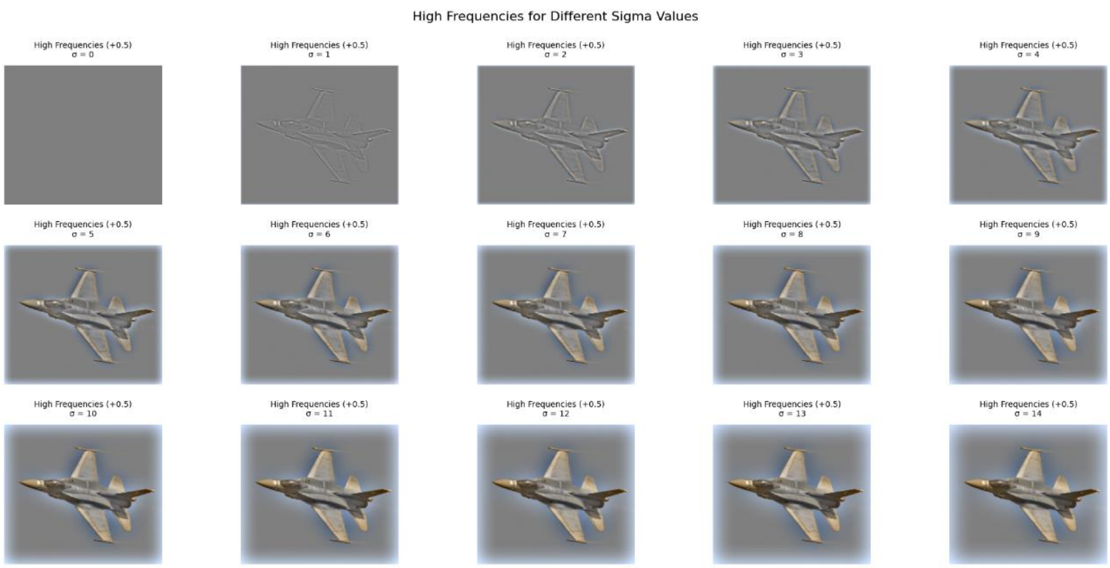

Fig.20 High frequencies component for different σ value

The base layer shows the complementary trend. In Fig.21, at $\sigma = 0$ the “low-pass” is simply the original bird. As σ increases, feather texture and small edge contrast fade, while silhouette and broad shading remain. By $\sigma = 5{\sim}7$ the bird becomes a genuinely blob-like figure suitable as a far-view percept. By $\sigma \geq 10$ it approaches a very soft blob; the corners darken because my "my_imfilter" uses zero padding—larger kernels give more weight to padded zeros near the borders, which creates a vignette.

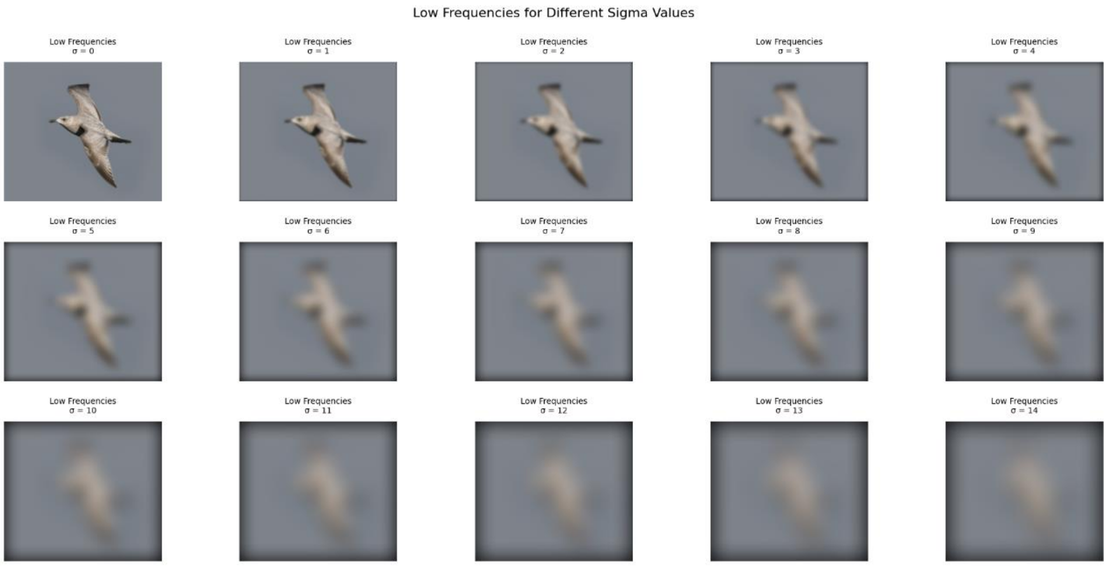

Fig.21 Low frequencies component for different σ value

The hybrids combine these two channels. Fig.22 shows that $\sigma = 0$ yields just the bird. At $\sigma = 1{\sim}2$, the plane appears as a light line drawing that does not yet carry enough shading to read clearly on top of a still-detailed bird; the far-view is also somewhat busy. In my sweep the most readable hybrid occurs at $\sigma = 3$: the plane's near-view identity is crisp and coherent—edges plus just enough mid-band structure—while halos are restrained and the background remains clean; at the same time the bird's coarse structure still dominates as I step back. As σ rises to $4\sim6$, the near-view plane grows stronger but with thicker halos and a faint hazy veil, and the far-view bird loses clarity as more mid-band energy from the plane leaks through. From $\sigma = 7$ upward the hybrid looks increasingly heavy: halos are obvious along leading edges and tips, and the far-view becomes progressively contaminated by residual detail from the plane, so the identity switch with distance is less decisive.

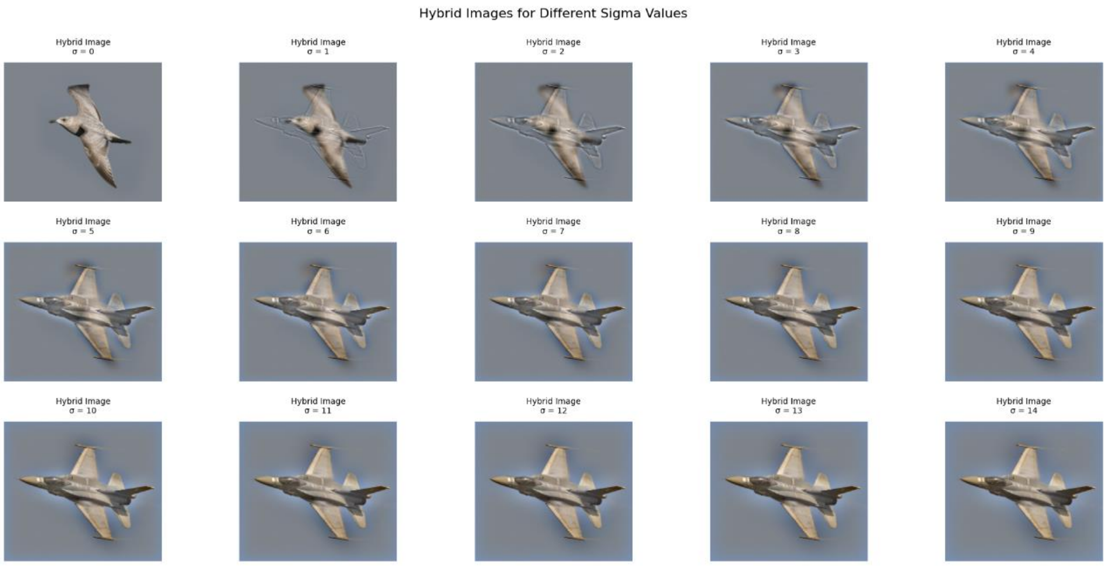

Fig.22 Hybrid images for different σ value

Runtime behavior is summarized in Fig.23. The average execution time increases steadily as σ goes from 0 to 14, with small local bumps. This is consistent with the fact that my Gaussian kernel size grows with σ (I use a spatial window on the order of $4\sigma + 1$) and every convolution sweeps that 2-D window over all pixels; larger σ means more multiply-accumulates per pixel and thus longer time. Minor non-monotonicities reflect fixed Python overheads and cache effects at nearby kernel sizes.

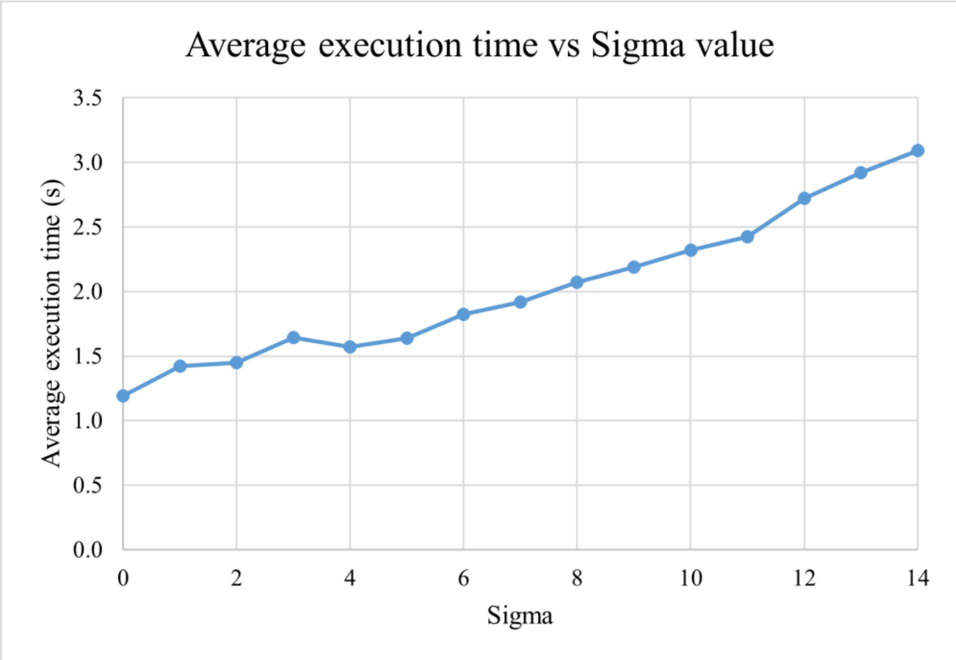

Fig.23 Average execution versus σ value

Why does σ control these trade-offs? The 1-D Gaussian kernel is $g(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\mathrm{e}^{\left(-\frac{x^2}{2\sigma^2}\right)}$. Its continuous-frequency magnitude response (with frequency f in cycles per pixel) is $H_\sigma(f) = e^{-2\pi^2\sigma^2 f^2}$, so increasing σ lowers the effective cutoff frequency. A convenient reference is the half-power point (−3 dB), where $H_\sigma(f_{-3\mathrm{dB}}) = \frac{1}{\sqrt{2}}$. Solving gives $f_{-3\mathrm{dB}} = \frac{\sqrt{\ln 2}}{2\pi\sigma} \approx \frac{0.133}{\sigma}$ cycles/pixel. As σ grows, $f_{-3\mathrm{dB}}$ shrinks: the low-pass gets blurrier, and the complementary high-pass widens from razor-thin edges (small σ) to thick, halo-prone relief (large σ). Visually, small σ leaves a skeletal near-view and a cluttered far-view; large σ yields heavy halos and an over-smoothed base with a vague far-view. At $\sigma = 3$ I get the best compromise for this pair: the high-pass contributes just enough mid-band shading to make the plane legible up close, the low-pass retains a clear bird silhouette at distance, halos remain controlled, and the runtime is moderate.

V. Alignment of two images before blending them together is important in implementing hybrid images. Try to create your own alignment flow for two arbitrary images. These two images should be in different scales. The sizes and the locations of the objects in the images that you want to blend should also be different. Please specify the steps and details in your flow. Show how to implement your own flow and show the results of hybrid images. Please put your source images in Lab06/data/ and your results in Lab06/results/. If you have any code, please put it in Lab06/code/.

Ans:

To ensure that hybrid images look convincing rather than “ghosted,” I align the two inputs before any frequency-based blending. I illustrate the process with the example pair in Fig.24 and Fig.25 and the resulting registered crops in Fig.26 and Fig.27; the multi-scale hybrid in Fig.28 only serves to demonstrate that the registration makes blending feasible.

My pipeline begins by converting each image to RGB floats and detecting 468 facial landmarks with MediaPipe Face Mesh in static-image mode. From the dense landmarks I extract five stable anchors in a fixed order—the outer corners of the left and right eyes, the nose tip, and the left and right mouth corners—so both faces are summarized by corresponding 2-D keypoints. Using these anchors, I build two candidate registrations. First, I assume no left–right flip and estimate an affine transform from the moving face to the reference face with RANSAC, which suppresses the influence of slightly misplaced landmarks. I then warp the moving image with bilinear interpolation and simultaneously warp a binary "all-ones" mask with nearest-neighbor interpolation to track which pixels remain valid after warping. Second, I allow for mirrored inputs: I horizontally flip the moving image and its five anchors about the image centerline, re-estimate an affine with RANSAC, and repeat the warping and mask generation.

Because affine warps introduce black padding outside the transformed footprint, I do not crop by a simple bounding box of valid pixels. Instead, I compute the intersection mask between the reference's full field and the warped moving mask, and I locate the largest axis-aligned rectangle that lies fully inside this intersection. Cropping both images to this "largest inscribed rectangle" removes the triangular black wedges that appear after rotation or shear and guarantees that every pixel in the crop is real for both views. Between the two candidates (with and without flip), I select the one whose inscribed rectangle has the larger area, on the assumption that better geometric agreement yields a larger clean overlap. The aligned results in Fig.26 and Fig.27 show that, despite the original differences in scale and framing (Fig.24–Fig.25), the eye corners, nose tip, and mouth corners are brought into correspondence inside a shared window with clean borders and identical dimensions. With this registration in place, I can directly apply the standard Gaussian low/high-pass blending (reported elsewhere); Fig.28 merely illustrates that the alignment enables a coherent hybrid across scales.

Fig.24 The first source image for alignment (168 * 299)

Fig.25 The second source image for alignment (275 * 183)

Fig.26, Fig.27 The result of alignment (168 * 173)

Fig.28 The hybrid image made by the aligned image pair

This alignment code is intentionally simple and has clear limitations. It is tailored to human faces because it relies on “MediaPipe Face Mesh”; it will not work on animals or arbitrary objects. Landmark detection can fail on extreme yaw/pitch, heavy occlusion (hair, hands, sunglasses), or very low resolution, and strong expressions may slightly shift the chosen five anchors. The affine model cannot completely compensate 3-D pose differences or perspective effects, so moderate residual misalignment may remain for non-frontal faces. The largest-inscribed-rectangle crop necessarily discards peripheral content to avoid padding, reducing the field of view.

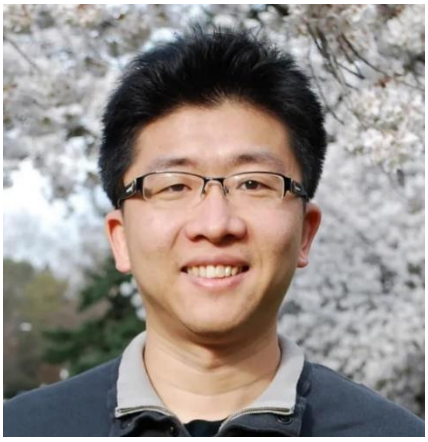

Fig.29, Fig.30 Another result of alignment (541 * 527)

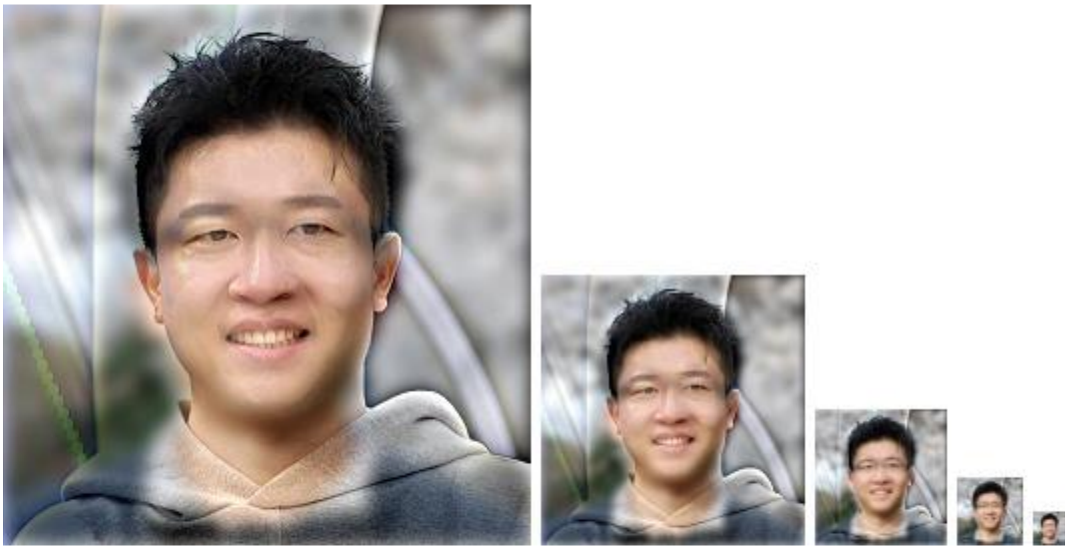

Fig.31 The hybrid image made by the aligned image pair

5. Conclusion

This lab deepened my understanding of spatial filtering and multiscale perception by taking hybrid images from first principles to a working system. I implemented a general-purpose convolution routine that supports RGB images and arbitrary odd-sized kernels, then used it to separate an image into low- and high-frequency bands and recombine them into hybrids. Through controlled comparisons I verified that a

Gaussian low-pass produces smoother, artifact-free bases than an equal-size box filter, which tends to leave blocky ringing; this directly translated to more natural hybrids. I also explored a reversed construction using a Laplacian high-pass and “low-pass by subtraction,” and observed that although this pipeline highlights edges well, its low-pass surrogate inherits haloing and contrast loss that reduce hybrid quality relative to the Gaussian pipeline.

A sigma sweep ($\sigma = 0 \ldots 14$) showed how cutoff frequency governs both the blur radius and what our visual system prioritizes at different viewing distances. In my results, $\sigma \approx 3$ balanced the low-frequency base and high-frequency details best for readability; very small σ left residual clutter from the high-pass image, while large σ over-blurred the base and dimmed the fused details. Execution time rose roughly monotonically with σ because larger kernels increase convolution cost, matching the separability/size trade-offs discussed in class. Finally, I built a robust face-alignment flow—landmark detection, five-point affine estimation with RANSAC, horizontal-flip fallback, warping, and mask-based intersection cropping—to normalize scale, rotation, and pose before blending. This alignment is crucial: when the facial anchors coincide, the two bands group perceptually rather than competing.

Overall, the lab demonstrates how (1) the choice of filter and its scale parameter directly shape the frequency content we present to the eye, (2) perceptually clean hybrids rely on well-behaved low-pass filtering and careful registration, and (3) engineering details such as kernel size, separability, and overlap-aware cropping matter as much as the high-level idea. Future improvements include exposure/color matching between sources, multi-band gains to compensate for contrast compression, and extending the alignment flow beyond faces to general objects with keypoint-based matching.

6. Reference

[1] Department of Electrical Engineering, National Tsing Hua University, *EE3662-2025-Unit II-Visual Signal*.

[2] Department of Electrical Engineering, National Tsing Hua University, *Lab6*.

[3] Aude Oliva, Antonio Torralba, and Philippe G. Schyns. 2006. Hybrid Images.

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